

# SciOdyssey.

## A research layer over your agent harness.

PRESERVE THE RESEARCH WORLD · RESET THE RESEARCHER

ROBUST

SUPERVISABLE

BROAD SEARCH

SAME EVIDENCE  
FRESH JUDGMENT  
A WIDER SEARCH

Research  
Runs  
Longer.

### 01 / PROBLEM

## Why current agents fall short.

Sustained autonomous research breaks in three different ways.

#### 01 Closed-world fragility

Unexpected runtime failures can halt an unattended workflow.

Cycle 1 · `numpy.float32` serialization error · autonomously repaired

#### 02 Trajectory momentum

A single long-running trajectory keeps moving in the same direction.

Direct Codex control reached ~**0.6046** after extended local refinement

#### 03 Context inflation cost

Reconstructing project context can cost far more than the edit itself.

Project-shaped replication · 340,087 vs 42,377 inclusive tokens · **8.03x**

#### DESIGN RESPONSE

Keep the world. Reset the trajectory.

### 02 / SYSTEM

## One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

#### DUAL-LIFETIME LOOP

Persistent world · ephemeral trajectory · audited write-back

#### Fresh Scientist

Owns scientific judgment inside one trajectory: hypotheses, code, experiments, interpretation, and pivots.

#### META Review

Audits claims against artifacts, scopes conclusions, and decides what is durable enough to survive.

#### SELECTIVE PERSISTENCE

Only audited empirical findings, code state, and bounded priors write back to the shared world.

#### PERSISTENT RESEARCH WORLD

What survives across trajectories

##### A · EVIDENCE

**Knowledge + Ledger**  
Metrics, failures, reports, and experiment receipts in `ledger.jsonl`.

##### B · PRIORS

**Brief + Hypotheses**  
Current understanding and exploration leads in `brief.md`, open to revision.

##### C · STATE

**Code + State**  
Retained implementation, serialized model weights, checks, and git provenance.

#### CONTEXT RESET

The next Scientist reads verified experience from disk, but does not inherit the previous reasoning trace.

#### RUNTIME SUBSTRATE

##### WORKTREE ISOLATION

Independent git worktrees isolate branches and preserve reproducible artifacts.

##### SELF-HEALING

Traceback → patch → rerun keeps unattended research moving after runtime faults.

##### PARALLEL SYNTHESIS

Completed branches can be compared post-hoc; useful ideas survive even when a branch loses.

The next Scientist inherits **verified experience** — not prior reasoning inertia.

Decoupled Memory · Audited Persistence · Fresh Scientific Judgment

### 03 / EVALUATION

## Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

#### 01 MEASURED RESULT

### The model got better.

**0.6016000**

Official FM baseline

→

**0.6059363**

SciOdyssey submission

**+0.0043363 Primary improvement**

GAUC **0.6728421**

(+0.0054)

nDCG@5 **0.5390304**

(+0.0033)

LogLoss **0.38102**

(-0.0032)

**Feature Engine:** 46 categorical cross-interactions + log1p transforms

**Architecture:** 8-seed Factorization Machine ensemble (NumPy / CPU)

**Monotonic gain** from baseline (0.6016) → feature engineering (0.6041) → seed ensembling (0.6059); **100% CPU inference** without GPU clusters.

#### 02 SUSTAINED SEARCH

### The search stayed productive.

**0.60163**

C1: Valid Base

**0.60409**

C2: Features

**0.60443**

C3: Var. Red.

**0.60594**

C4: 8-Seed FM

SciOdyssey **0.6059363** Climbed across 4 cycles (7 Full evals)  
Codex Goal 0.6044533 Peaked at C8; stalled & dropped in 22 runs

4 autonomous cycles

7 Full evaluations

**+0.00148** over Codex

**7 Full public-validation evaluations** retained across 4 cycles. Trajectory climbed via **structural feature expansion** rather than single lucky edits.

#### 04 RESOURCE ACCOUNTING

### We are not tokenmaxxing.

##### PROJECT COST

**~\$10**

First code to final result

##### HARDWARE

**1 × M2**

1 consumer laptop

##### COMPUTE

**0 GPU-hr**

100% CPU / NumPy only

##### AGENT RUNTIME

**1h 55m**

Wall-clock across 4 cycles

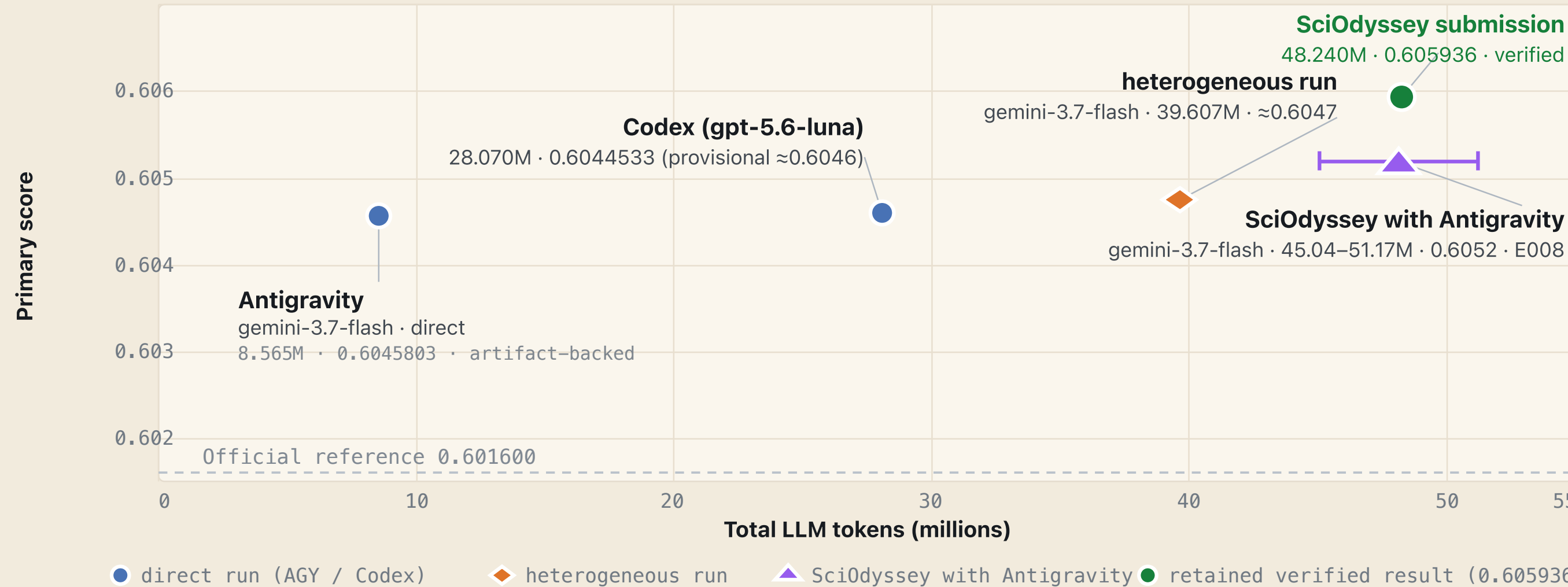
Performance came from the autonomous research loop, not an expensive compute budget. Full discovery executed on consumer hardware without GPU clusters.

#### 05 DIRECT AGENT BENCHMARK

### Score versus measured LLM-token investment.

KuaiRand-Pure Public Validation

Input + output including cache-read · Whole recorded agent system



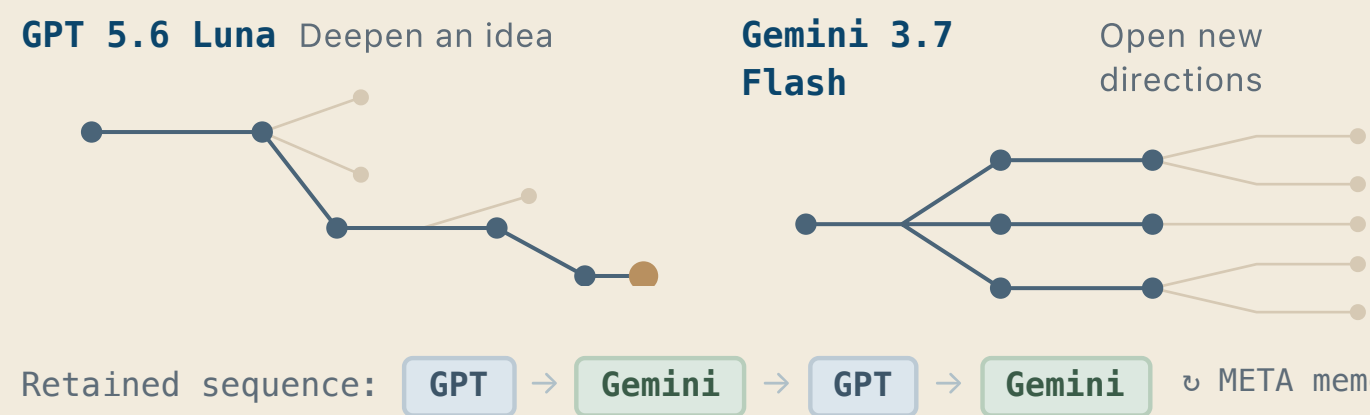
All results **verified** on KuaiRand-Pure official validation split with **complete reproducible artifacts**.

### 04 / INSIGHTS

## What the run taught us.

#### 01 · MODEL DIVERSITY

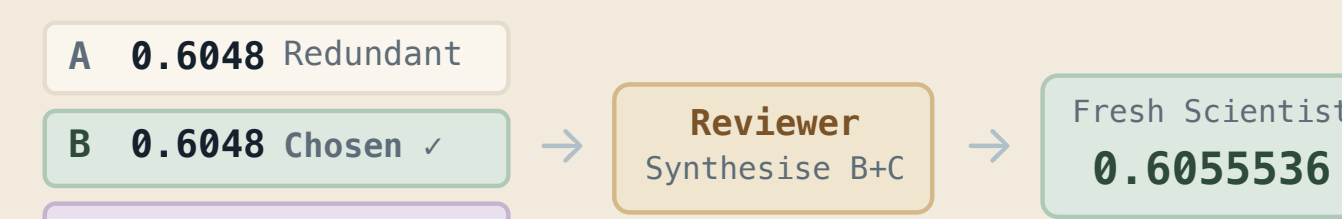
### Depth + Breadth Synergy.



Gemini-only **0.6052** → GPT+Gemini **0.6059363**

#### 02 · PARALLEL SYNTHESIS

### Cross-Branch Idea Transfer.



Lower-scoring Branch C contributed pairwise ranking and rank normalisation — a lower score can still carry a useful idea.

Branch B+C Ideas → Synthesis **0.6055536**

### 05 / EXPERIENCE

## Start your journey.

#### task.md

##### What to solve

**Formal task contract:** research objectives, dataset inputs, compute constraints, and official evaluation protocol.

- Dataset paths & split integrity
- Official sandbox evaluator

#### PERSONAL.md

##### How to work

**User research preferences,** model guidance, exploration style, tool permissions, and supervisory checkpoints.

- Model rotation (GPT / Gemini)
- Supervision & budget thresholds

#### SUPERVISION TIERS

**Tier 1** Unattended Overnight Runs · **Tier 2** Checkpoint Review (META) · **Tier 3** Steer Working Priors

1. Define task.md → 2. Single-step test → 3. Autonomous search → 4. Parallel explore → 5. Inspect evidence

#### DECLARATIVE PROTOCOL

Human defines boundary and preferences (task.md + PERSONAL.md); SciOdyssey autonomously iterates, heals, and crystallizes verified evidence.

